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Transportable machine including a track system

Abstract

A transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt, and a power supply provides power to the motor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a national phase filing under 35 U.S.C. 371 of International Application No. PCT/US2020/036914 filed Jun. 10, 2020, which claims priority to U.S. Provisional Patent Application No. 62/859,330 filed Jun. 10, 2019, U.S. Provisional Patent Application No. 62/924,533 filed Oct. 22, 2019, and U.S. Provisional Patent Application No. 62/987,383 filed Mar. 10, 2020, the entire contents of which are incorporated by reference herein.

FIELD OF INVENTION

(1) The present disclosure relates to a dolly or hand truck for transporting a load. Furthermore, the present disclosure relates to sewer cleaning machines used to remove or dislodge debris caught in a drain or pipe.

BACKGROUND

(2) Sewer cleaning machines are used to clean clogs and debris out of drains, sewers, and the like. Smaller handheld drain cleaners may be used to clean household drains from sinks or shower drains. However, larger and heavier cleaning machines are often used to clean sewers and industrial drains. A sewer cleaning machine may have as much as 200-300 feet of cable and a weight of 200-300 lbs. Accordingly, some sewer cleaning machines may be cumbersome to transport.

(3) Hand trucks are used to transport a load from a first location to a second location. Hand trucks may be able to transport relatively large loads. Accordingly, some hand trucks may be helpful in transporting large machines or tools, such as a sewer cleaning machine. However, hand trucks also have limitations. For example, hand trucks are primarily helpful when transporting a load on a relatively flat or a smooth surface, or when the load does not get overly heavy. However, a hand truck may be cumbersome to transport when the load becomes too large, or when the hand truck must be used to transport a load up a set of stairs.

SUMMARY

(4) In one embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller, an endless belt configured to rotate about the roller, and a belt tensioning assembly configured to tension the endless belt. The belt tensioning assembly includes a bracket supporting the roller, a mount coupled to the frame, and an adjustment member configured to adjust the position of the bracket relative to the mount. A motor is coupled to the roller, where the motor is operable to drive the endless belt.

(5) In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt, a power supply provides power to the motor, and a controller is coupled to the power supply and the motor. The controller is configured to monitor the power supply, determine when the power supply is insufficient to climb a set of stairs, and in response to determining that the power supply is insufficient to climb the set of stairs, send a signal to alert a user, to shutoff the motor, or both.

(6) In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, a wheel disposed on the bottom end of the frame, a track coupled to the frame, where the track includes a roller and an endless belt configured to rotate

about the roller, and a motor coupled to the roller. The motor is operable to drive the endless belt in a first direction at a first speed and in a second direction at a second speed, where the second speed is slower than the first speed.

(7) In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, a wheel disposed on the bottom end of the frame, and a track coupled to the frame. The track includes a roller and an endless belt configured to rotate about the roller, where the endless belt has a recess extending along an inside surface, and the recess receives a rib on the frame to align the endless belt relative to the frame. A motor is coupled to the roller, where the motor is operable to drive the endless belt.

(8) In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt. A sewer cleaning machine is supported by the base, where the sewer cleaning machine includes a drum and a cable positioned at least partially within the drum. A support system is configured to support the transportable machine on a surface and inhibits the transportable machine from tipping, where the support system includes one or more supports positioned below the base.

(9) In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. The track is movable relative to the elongated frame between a stored position and an extended position, where the track is positioned farther away from the frame when in the extended position than in the stored position. A motor is coupled to the roller, where the motor is operable to drive the endless belt.

(10) In another embodiment, the present disclosure provides a transportable machine including an elongated frame having a top end and a bottom end, a handle disposed on the top end of the frame, a base disposed on the bottom end of the frame, where the base and the frame together define a space configured to receive and support a load, and a wheel disposed on the bottom end of the frame. A track is coupled to the frame, where the track includes a roller and an endless belt configured to rotate about the roller. A motor is coupled to the roller, where the motor is operable to drive the endless belt. A lifting system is configured to tilt the base relative to a support surface.

(11) The above aspects may be implemented in different combinations and not necessarily in the ordered presented. Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front perspective view of a sewer cleaning machine.

(2) FIG. 2 is a side view of the transportable machine of FIG. 1.

(3) FIG. 3 is a rear perspective view of the transportable machine of FIG. 1 with a motor housing removed.

- (4) FIG. 4 is a cross-sectional view of the transportable machine of FIG. 1.
- (5) FIG. 5 is a side view of a frame of the transportable machine of FIG. 1.
- (6) FIG. 6 is a flow chart of a stair-drive check.
- (7) FIG. 7 is a perspective view of a transportable machine with a belt tensioning mechanism according to one embodiment.
- (8) FIG. 8 is a perspective view of the belt tensioning mechanism of FIG. 7.
- (9) FIG. 9 is a side view of the belt tensioning mechanism of FIG. 7.
- (10) FIG. 10 is an enlarged perspective view of the belt tensioning mechanism.
- (11) FIG. 11 is an enlarged side view of the belt tensioning mechanism of FIG. 10.
- (12) FIG. 12 is an exploded view of the belt tensioning mechanism of FIG. 10.
- (13) FIG. 13 is a front perspective view of a transportable machine according to another embodiment of the disclosure.
- (14) FIG. 14 is a rear perspective view of the transportable machine of FIG. 13 with belts removed.
- (15) FIG. 15 is an enlarged view of the transportable machine of FIG. 14.
- (16) FIG. 16 is a cross-sectional view of the transportable machine of FIG. 15 taken along lines 16-16.
- (17) FIG. 17 is a perspective view of a portion of the transportable machine of FIG. 13.
- (18) FIG. 18 is a side view of a portion of the transportable machine of FIG. 13.
- (19) FIG. 19 is a perspective view of a transportable machine, illustrating a track of the transportable machine in a stored position.
- (20) FIG. 20 is another perspective view of the transportable machine of FIG. 19, illustrating the track of the transportable machine in an extended position.
- (21) FIG. 21 is another perspective view of the transportable machine of FIG. 19.
- (22) FIG. 22 is a rear perspective view of the transportable machine of FIG. 19, illustrating an air bag member.
- (23) FIG. 23 is a partial perspective view of a base of the transportable machine of FIG. 21.
- (24) FIG. 24 is a perceptive view of the transportable machine of FIG. 19, illustrating movement of the transportable machine by a user relative to stairs.
- (25) FIG. 25 is a perspective view of another transportable machine embodying the disclosure, illustrating a handle of the transportable machine in a first position.
- (26) FIG. 26 is another perspective view of the transportable machine of FIG. 25, illustrating the handle in a second position.
- (27) FIG. 27 is a perspective view of yet another transportable machine embodying the disclosure, illustrating a securing mechanism.
- (28) FIG. 28 is a partial side view of the transportable machine of FIG. 27.

DETAILED DESCRIPTION

- (29) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.
- (30) FIGS. 1-4 illustrate a transportable machine according to one embodiment. The transportable machine illustrated in FIGS. 1-4 is a sewer cleaning machine **1200**. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the sewer cleaning machine **1200**.
- (31) The sewer cleaning machine **1200** includes a frame **1210**, a handle **1230**, a base **1234**, and a track **1226**. The frame **1210** defines an elongated body extending between the handle **1230** and the base **1234**. Specifically, the handle **1230** is positioned on a top end of the frame **1210** to enable a user to grip and control movement of the sewer cleaning machine **1200**. The base **1234** is

positioned on a bottom end of the frame **1210** and is configured to support a load, such as a machine or other load, that may be desirable to be able to transport. For example, the base **1234** may be configured to support many of the components of the sewer cleaning machine **1200**. In the embodiment illustrated in FIGS. **1-4**, the base **1234** supports a drum housing **1214**, a motor housing **1218**, and a power supply **1222**. The track **1226** is supported by the frame **1210** to help facilitate transport of the sewer cleaning machine **1200**. In the illustrated embodiment, the track **1226** is positioned on a rear side of the frame **1210** between the handle **1230** and the base **1234**.

Additionally, the illustrated sewer cleaning machine **1200** includes wheels **1238** rotatably coupled to the frame **1210** to also assist with transport of the sewer cleaning machine **1200**.

(32) In the illustrated embodiment, a drum **1242** is rotatably supported within the drum housing **1214** and includes a cable (not shown in FIGS. **1-4**) that is extendable out of an opening **1246** on the drum **1242**. In some embodiments, the drum **1242** is a cage-style drum that when the drum housing **1214** is opened allows easy access to the cable so a user can inspect the cable. The cable is extendable out of the drum with a cable feed device **1250**. A first motor **1254** is supported within the motor housing **1218** and is coupled to the drum **1242**. The first motor **1254** is operable to rotate the drum **1242**. Rotation of the drum **1242** creates friction between an inner surface of the drum **1242** and the cable, which causes the cable to spin to facilitate clearing debris from a drain pipe or another conduit.

(33) The track **1226** is configured to engage a surface, such as stairs or a ramp to help a user maneuver the sewer cleaning machine **1200**. In the illustrated embodiment, the track **1226** is positioned on the opposite side of the frame **1210** as the drum **1242**. For example, the drum **1242** is positioned on a front side of the frame **1210**, and the track **1226** is positioned on a back side of the frame **1210**. In other embodiments the track **1226** can be positioned on either the left or right side of the frame **1210**.

(34) The track **1226** includes a substantially horizontal drive shaft **1258** with a first drive roller **1262** at one end and a second drive roller **1266** at another end. A first endless belt **1270** extends around the first drive roller **1262** and a first idler roller **1274**, and a second endless belt **1278** extends around the second drive roller **1266** and a second idler roller **1282**. The endless belts **1270**, **1278** extend substantially vertically along a length of the frame **1210**. In some embodiments, the track **1226** may only include a first endless belt **1270** and respective rollers **1262**, **1274** rather than including first and second endless belts **1270**, **1278**. In some embodiments, the endless belts **1270**, **1278** include traction elements that assist in gripping a surface, ledge, or other object. For example, in the illustrated embodiment, the endless belts **1270**, **1278** include castellations **1276**, or projections, that help grip various surfaces. In further embodiments, the endless belts **1270**, **1278** include replaceable cleats that can be replaced when worn down instead of replacing the entire track **1226**. In other embodiments, the track **1226** can articulate or expand and retract to better climb stairs.

(35) As shown in FIG. **3**, a second motor **1286** is supported by the frame **1210** and is coupled to the drive shaft **1258** of the track **1226**. The second motor **1286** is operable to rotate the drive shaft **1258** and thus the drive rollers **1262**, **1266** to facilitate rotation of the endless belts **1270**, **1278**. In other embodiments, the track **1226** is passively driven rather than motor driven. For example, the endless belts **1270**, **1278** of the track **1226** may be rotated by engagement with a surface, such as stairs or a ramp, as a user pulls the sewer cleaning machine **1200**. Regardless of whether the track **1226** is motor driven or passively driven, the track **1226** is decoupled from the drum **1242** in that the track **1226** is independently rotated without affecting the rotation of the drum **1242**.

(36) In the illustrated embodiment, both the first and second motors **1254**, **1286** are powered by the power supply **1222** that is supported on the frame **1210**. The first and second motors **1254**, **1286** are, for example, brushless motors. In additional embodiments, the first and second motors **1254**, **1286** are variable two speed motors. In the illustrated embodiment, the power supply **1222** includes a battery receptacle that receives a battery pack to provide D/C power to the sewer cleaning

machine **1200**. For example, the battery receptacle may removably receive a rechargeable power tool battery pack. In further embodiments, the power supply **1222** may receive more than one battery pack to power the sewer cleaning machine **1200**. In alternative embodiments, the power supply **1222** may be coupled to a power outlet to provide A/C power to the sewer cleaning machine **1200**.

(37) The power supply **1222** is electrically coupled to a controller **1284** that may control operation of the first and second motors **1254**, **1286**. In some embodiments, the controller **1284** ensures that when one motor is operating the other motor is locked out and cannot be run. As previously mentioned, the track **1226** is decoupled from the drum **1242** such that rotation of one is independent of the other. In some embodiments, the controller **1284** actively decouples the track **1226** from the drum **1242** so that they cannot operate at the same time. In other embodiments, the track **1226** is decoupled from the drum **1242** only by lack of mechanical connection to the drum **1242**. Additionally, the sewer cleaning machine **1200** may include switches, buttons, a user interface, or other control features that allow a user to selectively control the sewer cleaning machine **1200**.

(38) Further, the power supply **1222** or the battery may include a battery fuel gauge to indicate to a user how much longer the battery will last. Additionally, the sewer cleaning machine **1200** may include battery detection that indicates to a user if the sewer cleaning machine **1200** has enough power to climb a standard set of stairs and, if not, lock out the tracks **1226** from being operated. This inhibits a user from initiating a stair climb and then running out of power in the middle of the stairs where it is difficult to manually maneuver the sewer cleaning machine **1200**.

(39) Referring to FIG. 6, a user can adjust the sewer cleaning machine **1200** from a drain cleaning mode, in which the drum **1242** is operable, to a stair climbing mode, in which the track **1226** is operable. When the user adjusts the sewer cleaning machine to the stair climbing mode, the control system of the sewer cleaning machine **1200** initiates a stair-drive check to determine whether the power supply **1222** has sufficient power to climb a set of stairs. When the remaining capacity of the power supply **1222** is sufficient to climb an entire set of stairs, the system will allow the second motor **1286** to power the track **1226** and enable the sewer cleaning machine **1200** to climb a set of stairs. Specifically, when the capacity of the power supply **1222** is above a predetermined threshold, the track **1226** is operable to climb the stairs. When the remaining capacity of the power supply **1222** is insufficient to climb an entire set of stairs, the system enters a lock out mode and prevents the second motor **1286** from powering the track **1226**. Specifically, when the capacity of the power supply **1222** is below a predetermined threshold, the track **1226** is locked out.

Alternatively, when the remaining power supply **1222** is insufficient to climb an entire set of stairs, the system provides an alert to the user indicating that the battery power is insufficient to climb the entire set of stairs. In this embodiment, the user may have the option to continue to utilize the track **1226** in a driven mode until the power supply **1222** is completely depleted and then may climb the remaining stairs using the track **1226** in an idle mode.

(40) The predetermined threshold can be determined based on a standard height of a set of stairs. In some embodiments, the predetermined threshold can be determined based on a variety of different sized stair sets, such as a half stair well or an extra long stair well. When the sewer cleaning machine **1200** enters the lock out mode, an indicator informs the user that there is insufficient battery to complete the stair climb. The indicator may include, for example, a light on the sewer cleaning machine **1200** and/or an audible alert. The user may then charge the power supply **1222** or swap battery packs so that the capacity exceeds the predetermined threshold. When the sewer cleaning machine **1200** is in the locked out mode, a user has the option to adjust the sewer cleaning machine back to a drain cleaning mode. Although the power supply may not be sufficient to initiate a stair climb, the drum **1242** may still be operated when the capacity of the power supply **1222** is below the predetermined threshold.

(41) Referring back to FIGS. 2-4, the sewer cleaning machine **1200** is shown in a first operational

position. In this position, the drum housing **1214** of the sewer cleaning machine **1200** is supported on a surface to facilitate the clearing of debris from a conduit. In operation, the power supply **1222** supplies power to the first motor **1254** to spin the drum **1242**. The cable feed device **1250** draws cable from inside the drum **1242** so that a user may extend the cable into a drain. Rotation of the drum **1242** causes the cable to spin, assisting in the removal of debris from the drain.

(42) With reference to FIGS. **1** and **5**, the sewer cleaning machine **1200** is shown in a second transport position. A user may tilt the frame **1210**, lifting the housing **1214** off of a surface and allowing the wheels **1238** to transport the sewer cleaning machine **1200** along the surface. However, due to the weight, the sewer cleaning machine **1200** may be difficult to lift. Specifically, a user may have difficulty transporting the sewer cleaning machine **1200** on stairs. During transportation, the track **1226** may assist in lifting the sewer cleaning machine **1200** both up and down stairs. A user may first position the sewer cleaning device **1200** so that the track **1226** engages the stairs, for example as shown in FIG. **5**. Once the track **1226** engages the stairs, the user can control the power supply **1222** to operate the second motor **1286**. The second motor **1286** rotates the drive shaft **1258**, thereby rotating the endless belts **1270**, **1278**. As the endless belts rotate **1270**, **1278**, the traction on the belts **1270**, **1278** assist in pulling the sewer cleaning machine **1200** up the stairs. Meanwhile, the user can also assist by pulling on the handle **1230** of the sewer cleaning device **1200**. To transport the sewer cleaning device **1200** down stairs, a user can control the second motor **1286** to rotate the track **1226** in the opposite direction.

(43) With continued reference to FIG. **5**, the center of gravity X.sub.c of the sewer cleaning machine **1200** is positioned on the track **1226** while the sewer cleaning machine **1200** is at an angle relative to the ground (i.e., horizontal). In some embodiments, the angle of the sewer cleaning machine **1200** is between approximately 30 and 35 degrees with the horizontal. In other embodiments, the angle of the sewer cleaning machine **1200** is between approximately 32 and 33 degrees with the horizontal. In further embodiments, the angle of the sewer cleaning machine **1200** is 32.47 degrees with the horizontal, which correlates with the standard US code for the angle of stairs. Positioning the center of gravity X.sub.c on the treads reduces the possibility of the sewer cleaning machine **1200** tipping if a user were to release the sewer cleaning machine **1200** while climbing or descending stairs. In some embodiments, when the sewer cleaning machine **1200** is tilted at an angle less than 35 degrees relative to the horizontal, the center of gravity X.sub.c of the sewer cleaning machine may be above the tracks in a vertical direction. In a further embodiment, when the sewer cleaning machine **1200** is tilted at an angle of less than 33 degrees, the center of gravity X.sub.c is above the treads in a vertical direction.

(44) In some embodiments, the sewer cleaning machine **1200** includes a variable speed control that allows for stair climbing at a user selectable speed. For example, the sewer cleaning machine **1200** may travel up a set of stairs at a different speed than traveling down a set of stairs. In some embodiments, the sewer cleaning machine **1200** may travel up the set of stairs faster than the machine **1200** travels down the set of stairs. The user may adjust the speed of the track **1226** to travel up or down a set of stairs at a desired speed. In some embodiments, the controller **1284** is able to automatically control the speed of the track **1226** based on a direction of travel of the sewer cleaning machine **1200**. For example, the controller **1284** may determine whether the sewer cleaning machine **1200** is traveling up or down a set of stairs and automatically determine and adjust the desired speed of the track **1226**. In some embodiments, the controller **1284** may determine when the track **1226** is rotating a first direction (e.g., clockwise) correlated with traveling up a set of stairs, and thereby increase the speed of the track **1226**. Likewise, the controller **1284** may determine when the track **1226** is rotating in a second direction (e.g., counterclockwise) correlated with traveling down a set of stairs, and thereby decrease the speed of the track **1226**. In some embodiments, the user may control or may override the speed of the track **1226** designated by the controller **1284**.

(45) With reference to FIGS. **7-12**, the track **1226** may include a belt tensioning assembly **1290**. In

the illustrated embodiment, the track **1226** includes two belt tensioning assemblies **1290**, one of each endless belt **1270**, **1278**. In other embodiments, the track **1226** may include fewer or more belt tensioning assemblies **1290** (depending on the number of belts), or a single belt tensioning assembly may be used for multiple belts. The belt tensioning assemblies **1290** can increase or decrease the tensions of the first endless belt **1270** and the second endless belt **1278**. For example, increasing the tension of the belts **1270**, **1278** may reduce the possibility of the belts **1270**, **1278** slipping during rotation. Increasing the tension of the belts **1270**, **1278** may also reduce the possibility of the belts **1270**, **1278** derailing from the pulley system. In addition, when the belts **1270**, **1278** become worn and need to be replaced, the tension of the belts **1270**, **1278** may be decreased in order to more easily remove the old belt and reattach a new belt. One of the belt tensioning assemblies **1290** is described in more detail below. The other belt tensioning assembly **1290** is substantially the same.

(46) The illustrated belt tensioning assembly **1290** includes a bracket **1294**, a mount **1298**, and an adjustment member **1300**. The bracket **1294** is movable relative to the mount **1298**. The mount **1298** supports the adjustment member **1300**, which helps to position the bracket **1294** relative to the mount **1298**. The belt tensioning assembly **1290** is capable of adjusting the position of an idler roller, such as the first idler roller **1274** and the second idler roller **1282** relative to the first endless belt **1270** and the second endless belt **1278**, respectively. As described herein, the belt tensioning assembly **1290** will be described with respect to the first idler roller **1274** and first endless belt **1270**, however, it should be understood that the same construction can be applied to the second idler roller **1282** and the second endless belt **1278**.

(47) The bracket **1294** supports the first idler roller **1274**. In particular, the bracket **1294** rotatably supports the first idler roller **1274** such that the first idler roller **1274** is rotatable relative to the bracket **1294**. However, the bracket **1294** and the first idler roller **1274** translate together relative to the first endless belt **1270**. As such, the bracket **1294** and the first idler roller **1274** can be translated in a first direction **1310** to increase the tension of the first endless belt **1270**, and can be translated in a second direction **1312** to decrease the tension of the first endless belt **1270**.

(48) The mount **1298** is fixed to a portion of the frame **1210**. The bracket **1294**, and thus the first idler roller **1274**, can translate relative to the mount **1298** in order to tension the first endless belt **1270**. Specifically, the adjustment member **1300** is configured to adjust the position of the bracket **1294** and the first idler roller **1274** relative to the mount **1298**. In the illustrated embodiment, the adjustment member **1300** is a threaded bolt **1304** and a nut **1308**; however, in other embodiments, different types of adjustment members may be used. The nut **1308** can be threaded along the length of the bolt **1304** to adjust the position of the bracket **1294**. For example, the bolt **1304** can be threaded in the first direction **1310** along the bolt **1304** to move the bracket **1294** in the first direction **1310** to increase tension in the first idler belt **1270**. Similarly, the bolt **1304** can be threaded in the second direction **1312** along the bolt **1304** to move the bracket **1294** in the second direction **1312** to decrease tension in the first idler belt **1270**. In addition, in the illustrated embodiment, the mount **1298** includes a pair of posts **1313** that extend through slots **1314** in the bracket **1294** to help align and guide the bracket **1294** as it moves relative to the mount **1298**.

(49) In the illustrated embodiment, the nut **1308** can be accessed from both sides of the first endless belt **1270** through a window **1318** formed by the mount **1298**. This allows a user to use a wrench to thread the nut **1308** and adjust the tension of the first endless belt **1270**, thus reducing the amount of force input by the user.

(50) FIG. 13 illustrates a transportable machine according to another embodiment. The transportable machine illustrated in FIG. 13 is a sewer cleaning machine **1400**. The sewer cleaning machine **1400** is similar to the sewer cleaning machine **1200** with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the sewer cleaning machine **1400**.

(51) The sewer cleaning machine **1400** includes the frame **1410**, a handle **1430**, a base **1434**, and a track **1426**. The frame **1410** defines an elongated body extending between the handle **1430** and the base **1434**. Specifically, the handle **1430** is positioned on a top end of the frame **1410** to enable a user to grip and control movement of the sewer cleaning machine **1400**. The base **1434** is positioned on a bottom end of the frame **1410** and is configured to support a load, such as a machine or other load, that may be desirable to be able to transport. For example, the base **1434** may be configured to support many of the components of the sewer cleaning machine **1400**. In the illustrated embodiment, the base **1434** supports a drum housing **1414**, a motor housing **1418**, a power supply **1422**, and a track **1426** that is similar to the track **1226**. The track **1426** is supported by the frame **1410** to help facilitate transport of the sewer cleaning machine **1400**. In the illustrated embodiment, the track **1426** is positioned on a rear side of the frame **1410** between the handle **1430** and the base **1434**. Additionally, the illustrated sewer cleaning machine **1400** includes wheels rotatably coupled to the frame **1410** to assist in transport of the sewer cleaning machine **1400**.

(52) Moving to FIG. **14**, the track **1426** includes a first extrusion **1424** coupled to the frame **1410** and a second extrusion **1428** coupled to the frame **1410** on the opposite side of the drive shaft **1458** as the first extrusion **1424**. The first and second extrusions **1424**, **1428** each respectively support the first and second drive rollers **1462**, **1466**, the first and second idler rollers **1474**, **1482**, the first and second endless belts **1470**, **1478** (FIG. **16**), and the belt tensioning assemblies **1490**. Each of the first and second extrusions **1424**, **1428** are coupled to frame **1410** with a bracket **1438** (FIG. **17**). The bracket **1438** couples an upper portion of the first and second extrusions **1424**, **1428** to the frame **1410**. The brackets **1438** are coupled to both the frame **1410** and first and second extrusions **1424**, **1428** using fasteners **1450** (FIG. **17**) that extend through fastener slots.

(53) With reference to FIGS. **15** and **16**, both the first and second extrusions **1424**, **1428** include ribs **1436** that extend the length of the first and second extrusions **1424**, **1428** between the idler rollers **1474**, **1482** and the drive rollers **1462**, **1466**. In the illustrated embodiment, a first rib **1436** is on a front side of the first and second extrusions **1424**, **1428** facing the frame **1410** and a second rib **1436** is on a back side of the first and second extrusions **1424**, **1428** facing away from the frame **1410**. The ribs **1436** are generally placed in the center of the first and second extrusions **1424**, **1428**. The ribs **1436** each correspond to a recess **1440** on the inside of the endless belts **1470**, **1478**. The ribs **1436** hold the belts **1470**, **1478** in place while the belts **1470**, **1478** are being rotated. In the illustrated embodiment, each rib **1436** is a continuous projection extending along the length of the corresponding extrusion **1424**, **1428**. In other embodiments, the ribs **1436** may include discrete projections that only extend along portions of the length of the corresponding extrusion **1424**, **1428**. In other embodiments, the rib **1436** may only be on one side of the first and second extrusions **1424**, **1428**. For example, the rib **1436** may only be on a side facing away from the frame **1410** or only on a side facing the frame **1410**.

(54) As seen in FIG. **17**, the track **1426** further includes a bracket **1438** coupled to the outside of each of the first and second extrusions **1424**, **1428**. The bracket **1438** includes a flange portion **1442** and a circular portion **1446**. The flange portion **1442** includes fastener holes for fasteners **1450** to extend through and couple the bracket **1438** to the respective extrusion **1424**, **1428**. The flange portion **1442** extends past the respective extrusion **1424**, **1428** towards the frame **1410**. The portion of the flange portion **1442** that extends past the first and second extrusions **1424**, **1428** keeps the belts **1470**, **1478** from slipping or jumping off the first and second extrusions **1424**, **1428** during operation of the track **1426**. Each circular portion **1446** includes an opening **1454** adjacent the respective drive rollers **1462**, **1466** to protect the drive rollers **1462**, **1466** and provide support to the drive rollers **1462**, **1466** during operation of the track **1426**.

(55) With continued reference to FIG. **17**, an inner support **1460**, or guide block, is supported on the inside of each of the first and second extrusions **1424**, **1428**. The fasteners **1450** used to couple the brackets **1438** to the first and second extrusions **1424**, **1428** are also used to support the inner supports **1460** within the first and second extrusions **1424**, **1428**. Each inner support **1460** includes

an outer surface **1464** that extends from within the first and second extrusions **1424**, **1428**. The outer surface **1464** is generally aligned with or flush with an adjacent outer surface of the corresponding extrusion **1424**, **1428**. The outer surface **1464** assists the belts **1470**, **1478** in transitioning from the first and second extrusions **1424**, **1428** to the cleats of the drive rollers **1462**, **1466**. An inner surface of the inner support **1460** is curved or arcuate to match the curvature of the corresponding drive roller **1462**, **1466**. As such, the inner support **1460** can be positioned close to, but not interfere with the corresponding driver roller **1462**, **1466**.

(56) With reference to FIGS. **13**, **14**, and **18**, the sewer cleaning machine **1400** further includes a support system **1406**. The support system **1406** helps stabilize the sewer cleaning machine **1400** when in the upright position. The support system **1406** includes a first stationary kickstand **1408**, a second stationary kickstand **1412**, and a tilt block **1416** (FIG. **14**). The first and second kickstands **1408**, **1412** and the tilt block **1416** are generally supported beneath the frame **1410** when the sewer cleaning machine **1400** is in the upright position. In some embodiments, the support system **1406** may be considered part of the frame **1410** of the sewer cleaning machine **1400**. The first and second kickstands **1408**, **1412** are generally tubular and are fixed relative to the frame **1410**. The first kickstand **1408** is generally positioned in front of the drum housing **1414** underneath the feed device **1250**. The second kickstand **1412** is positioned behind the drum housing **1414**. Both the first and second kickstands **1408**, **1412** include rubber ends **1420** to inhibit the sewer cleaning machine **1400** from moving when in the upright position. The tilt block **1416** is positioned behind the second kickstand **1412** relative to the drum housing **1414**. The tilt block **1416** may be solid to add weight to the support system **1406**. The tilt block **1416** inhibits the sewer cleaning machine **1400** from tipping backwards when a cable is removed from the drum housing **1414** and the sewer cleaning machine **1400** is back heavy.

(57) In some embodiments, the sewer cleaning machine **1400** may include a pop-up wheel or bracket. The pop-up wheel or bracket may be activated to inhibit the sewer cleaning machine from falling down a flight of stairs if a user were to accidentally lose control or hold of the sewer cleaning machine **1400** while climbing stairs. Additionally, the pop-up wheel or bracket may be extended in between climbing steps of a flight of stairs to help support the sewer cleaning machine and inhibit the sewer cleaning machine from tipping backwards or forward. The pop-up wheel may be activated either manually or with a spring release to extend from under the frame **1410** of the sewer cleaning machine **1400**. In this way, when there is a pause in stair climbing the frame **1201** would rest on the pop-up bar or bracket that is supported by the previous step.

(58) FIGS. **19-24** illustrate a transportable machine according to another embodiment of the disclosure. The transportable machine illustrated in FIGS. **19-24** is a dolly or hand truck **2010**. The hand truck **2010** includes structure similar to the sewer cleaning machines **1200**, **1400** of the previous embodiments with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the hand truck **2010**.

(59) The hand truck **2010** includes a frame **2210**, a handle **2230**, a base **2234**, and a track **2226**. The frame **2210** defines an elongated body extending between the handle **1430** and the base **2234**. Specifically, the handle **1430** is positioned on a top end of the frame **2210** to enable a user to grip and control movement of the hand truck **2010**. The base **2234** is positioned on a bottom end of the frame **2210** and is configured to support a load, such as a machine or other load, that may be desirable to be able to transport. For example, the base **2234** may be configured to support a machine, tools, cargo, building materials, or other heavy items that may be difficult to transport. In the illustrated embodiment, the base **2234** includes a loading plate member **2040**. The loading plate member **2040** and the frame **2210** cooperatively define a space **2044** configured to receive and support a load (e.g., boxes, furniture, appliances, power tools, other goods, and the like) on the hand truck **2010**. The track **2226** is supported by the frame **2210** to help facilitate transport of the

hand truck **2010**. In the illustrated embodiment, the track **2226** is positioned on a rear side of the frame **2210** between the handle **2230** and the base **2234**. Additionally, the illustrated hand truck **2010** includes wheels **2238** rotatably coupled to the frame **2210** to assist in transport of the hand truck **2010**.

(60) The track **2226** is configured to engage a surface, such as stairs or a ramp to help a user maneuver the hand truck **2010**. In the illustrated embodiment, the track **2226** is positioned on the opposite side of the frame **2210** as the space **2044**. For example, the space **2044** is defined on a front side of the frame **2210**, and the track **2226** is positioned on a back side of the frame **2210**. The track **2226** includes a substantially horizontal drive shaft having a first endless belt **2270** that extends around a first drive roller and a first idler roller, and a second endless belt **2278** that extends around the second drive roller and a second idler roller. The endless belts **2270**, **2278** extend substantially vertically along a length of the frame **2210**. A motor **2286** (FIG. 20) is supported by the frame **2210** and is coupled to the drive shaft of the track **2226**. The motor **2286** is operable to rotate the drive shaft and thus the drive rollers to facilitate rotation of the endless belts **2270**, **2278**. The motor **2286** is powered by a power supply **2222** that is supported on the frame **2210**. In some embodiments, the endless belts **2270**, **2278** include traction elements that assists in gripping a surface, ledge, or other object.

(61) A user may tilt the frame **2210**, lifting the loading plate member **2040** off of a surface and allowing the wheels **2238** to transport the hand truck **2010** along the surface. However, due to the weight, the hand truck **2010** may be difficult to lift. Specifically, a user may have difficulty transporting the hand truck **2010** on stairs or into a truck bed. During transportation, the track **2226** may assist in lifting the hand truck **2010** both up and down stairs. A user may first position the hand truck **2010** so that the track **2226** engages the stairs, for example as shown in FIG. 24. Once the track **2226** engages the stairs, the user can control the power supply **2222** to operate the motor **2286**. The motor **2286** rotates the endless belts **2270**, **2278**. As the endless belts rotate **2270**, **2278**, the traction on the belts **2270**, **2278** assist in pulling the hand truck **2010** up the stairs. Meanwhile, the user can also assist by pulling on the handle **2230** of the hand truck **2010**. To transport the hand truck **2010** down stairs, a user can control the motor **2286** to rotate the track **2226** in the opposite direction.

(62) In some embodiments, the track **2226** can articulate or expand and retract relative to the frame **2210** to better climb stairs. For example, FIG. 19 illustrates the hand truck **2010** in a first (stored) position. In this position, the loading plate member **2040** of the base **2234** of the hand truck **2010** is supported on a surface to facilitate supporting of the load. FIGS. 20 and 24 illustrate the hand truck **2010** in a second, extended (operational) position. In the second position, the track **2226** is moved or pivoted away from the frame **2210**. In the illustrated embodiment, the track **2226** includes an axle positioned proximate the base **2234**, and the track **2226** is movable (e.g., pivotable) between the first position and the second position about the axle. The illustrated hand truck **2010** further includes a lock mechanism **2048** for maintaining the track **2226** in the second position. The illustrated lock mechanism **2048** includes a sub frame **2052** movably coupled to the frame **2210**, and a latch **2056** (i.e., handle latch) selectively coupled to the sub frame **2052**. The latch **2056** may be operated by a user using one hand. In particular, when the hand truck **2010** is in the second, extended position, the wheels **2238** are tilted away from the stairs such that the only the track **2226** is configured to contact the stairs.

(63) With reference to FIGS. 21-23, the hand truck **2010** further includes a lifting system **2060** to facilitate tilting of the hand truck **2010** toward or away from a surface. In the illustrated embodiment, as shown in FIGS. 21-23, the lifting system **2060** includes an air bag member **2064** supported by the base **2234**. More specifically, the illustrated air bag member **2064** is positioned between the loading plate member **2040** and a lifting plate member **2068** of the base **2234**. The lifting system **2060** further includes a pump **2072** and a foot pedal member **2076** supported by the frame **2210** and track **2226**. The pump **2072** is electrically coupled to the power supply **2222**. The

illustrated pump **2072** and foot pedal **2076** are positioned proximate the back side of the frame **2210**. The air bag member **2064** is configured to be inflated such that the loading plate member **2040** moves or pivots relative to the lifting plate member **2068**. The inflation of the air bag member **2064** adjusts the hand truck **2010** from a vertical or loading state to a tilted state (FIG. 23). A release member **2080** is provided proximate the handle **2230** for releasing or deflating the air bag member **2064** from the tilted state to the loading state. The release member **2080** includes an actuator (e.g., lever) operable by a user. In other embodiments, instead of the air bag member **2064**, the lifting system **2060** includes a powered actuator such as a gas cylinder, a piston, or other mechanical means, to pivot the loading plate member **2040** relative to the lifting plate member **2068**. Still further, in other embodiments, the lifting plate member **2068** may include a wheel coupled thereto to facilitate guiding of the hand truck **2010** on a flat surface. The lifting system **2060** is configured to facilitate tilting of the hand truck **2010** toward the track **2226**, thereby reducing the need to manual tilt the hand truck **2010** toward the track **2226**.

(64) The handle **2230** includes a control system having a plurality of switches **2083** positioned on the handle **2230** and a control panel **2084** positioned proximate a top side of the hand truck **2010**. The control system includes a controller electrically connected to the motor **2286** for controlling operation of the hand truck **2010**. In particular, the plurality of switches **2083** on the handle **2230** includes an actuator (e.g., trigger) for controlling operation of the track **2226**. In the illustrated embodiment, the switches **2083** include a first switch for turning on and off the motor **2286**. Still further, in other embodiments, the handle **2230** may also contain controls for additional functions of the stair climbing hand truck **2010** (e.g., variable speed). The illustrated control panel **2084** is positioned on the top side of the stair climbing hand truck **2010** to facilitate access of the user to the control panel **2084** for changing settings, features and other functions of the hand truck **2010**.

(65) FIGS. 25-26 illustrate a dolly or hand truck **2110** according to another embodiment of the disclosure. The hand truck **2110** includes structure similar to the hand truck **2010** with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the hand truck **2110**.

(66) In this embodiment, the handle **2230'** is movable between a first, angled position (FIG. 25) and a second, extended position (FIG. 26). In the illustrated embodiment, the handle **2230'** is pivotable about a pin **2088**. The handle **2230'** further includes an adjustment mechanism for adjusting the handle **2230'** between the first position and the second position. The illustrated adjustment mechanism includes an actuator **2092** (e.g., pad) operatively coupled to the pin **2088**. A user may actuate the actuator **2092** (e.g., push) for disengaging the pin **2088** from a detent mechanism, thereby adjusting the handle **2230'** between the first and second positions. The first position may facilitate movement of the hand truck **2110** on a flat surface by a user. The second position may facilitate movement of the hand truck **2110** on stairs by a user.

(67) FIGS. 27-28 illustrates a dolly or hand truck **2310** according to another embodiment of the disclosure with like features being represented by like reference numerals. As will be understood, the features shown and described herein with respect to the other embodiments of a transportable machine may be used in combination with the features shown and described with respect to the hand truck **2310**.

(68) The hand truck **2310** includes structure similar to the hand truck **2010** with like features being represented by like reference numerals. In this embodiment, the hand truck **2310** includes a securing mechanism **2096** to facilitate securing of a load **3000** received in the space **2044** defined by the hand truck **2310**. In the illustrated embodiment, the hand truck **2310** includes a plurality of adjustable straps **2096A-C**. In some embodiments, the adjustable straps **2096A-C** are ratchet straps. Additionally or alternatively, the adjustable straps **2096A-C** may be retractable straps. Each of the straps **2096A-C** is supported by the frame **2210**. In particular, each strap **2096A-C** extends from a first frame member **2210A** of the frame **2210** to a second frame member **2210B** in a lateral

direction relative to the space **2044** configured to receive the load **3000**. Each strap **2096A-C** is adjustable to individually lengthen or shorten each strap **2096A-C**. In some embodiments, one or more of the straps **2096A-C** is automatically retractable. In the illustrated embodiment, the securing mechanism **2096** includes three straps **2096A-C**. In other embodiments, the hand truck **2310** may include one or more straps. As shown in FIG. **28**, the ratchet mechanism of each adjustable strap **2096A-C** is secured to a support member **3004** (e.g., loop) positioned on the first frame member **2210A**.

(69) Although the disclosure is described with reference to discrete embodiments, variations of the sewer cleaning machines and hand trucks exist within the spirit and scope of the disclosure. The features shown and described herein with respect to one of the embodiments of a transportable machine may be used in combination with any of the features shown and described with respect to another embodiment of a transportable machine. For example, the following features may be used in any combination with one another, including but not limited to, the belt tensioning assembly (e.g., shown in FIGS. **7-12**), the stands (e.g., shown in FIG. **18**), the movable track assembly (e.g., shown in FIG. **20**), the lifting assembly (e.g., shown in FIG. **21-23**), or the movable handle (e.g., shown in FIGS. **25-26**). Additionally, any of the aforementioned features may be used in combination with the various control features, including but not limited to, the lock out system, which prevents a machine motor from being operated simultaneously with the track motor, a power monitoring system, which warns a user or shuts down the track motor when the power is insufficient to climb a full set of stairs, or a variable speed control system that operates the track at different speeds depending on whether the transportable machine is climbing up or down a set of stairs.

Claims

1. A transportable machine, comprising: an elongated frame having a top end and a bottom end; a handle disposed on the top end of the frame; a base disposed on the bottom end of the frame, the base and the frame together defining a space configured to receive and support a load; a wheel disposed on the bottom end of the frame; a track coupled to the frame, the track including a first roller, a second roller, an endless belt configured to rotate about the first and second rollers, and a belt tensioning assembly configured to tension the endless belt, the belt tensioning assembly including a bracket supporting the first roller, a mount coupled to the frame, and a single adjustment member accessible on each side of the endless belt and configured to adjust the position of the bracket relative to the mount to adjust a tension of the endless belt; and a motor coupled to the second roller, the motor operable to drive the endless belt.
2. The transportable machine of claim 1, wherein the endless belt includes a recess extending along an inside surface, the recess receiving a rib on the frame to align the endless belt relative to the frame.
3. The transportable machine of claim 1, wherein the track further includes a third roller, a fourth roller, and a second endless belt configured to rotate about the third and fourth rollers.
4. The transportable machine of claim 1, further comprising a sewer cleaning machine supported by the base, the sewer cleaning machine including a drum, a cable positioned at least partially within the drum, and a second motor driving rotation of the drum.
5. The transportable machine of claim 1, wherein the handle is movable relative to the elongated frame between an angled position, in which the handle extends generally downward from the top end of the frame, and an extended position, in which the handle extends generally upward from the top end of the frame.
6. The transportable machine of claim 1, wherein the track is movable relative to the elongated frame between a stored position and an extended position, wherein the track is positioned farther away from the frame when in the extended position than in the stored position.

7. The transportable machine of claim 1, wherein the single adjustment member is positioned between the first roller and the second roller.
8. A transportable machine, comprising: an elongated frame having a top end and a bottom end; a handle disposed on the top end of the frame; a base disposed on the bottom end of the frame, the base and the frame together defining a space configured to receive and support a load; a wheel disposed on the bottom end of the frame; a track coupled to the frame, the track including a first roller, a second roller, a frame member extending between the first and second rollers, and an endless belt configured to rotate about the first and second rollers, the endless belt having a recess formed on an inside surface of the endless belt, the recess receiving a rib on the frame member to align the endless belt relative to the frame member; and a motor coupled to the second roller, the motor operable to drive the endless belt.
9. The transportable machine of claim 8, wherein the track further includes a bracket positioned adjacent the endless belt and configured to inhibit the endless belt from slipping off of the frame.
10. The transportable machine of claim 8, wherein the track further includes a third roller, a fourth roller, and a second endless belt configured to rotate about the third and fourth rollers.
11. The transportable machine of claim 8, further comprising a belt tensioning assembly configured to tension the endless belt.
12. The transportable machine of claim 8, further comprising a sewer cleaning machine supported by the base, the sewer cleaning machine including a drum, a cable positioned at least partially within the drum, and a second motor driving rotation of the drum.
13. The transportable machine of claim 12, wherein operation of one of the first motor and the second motor prevents operation of the other of the first motor and the second motor.
14. The transportable machine of claim 8, wherein the recess also receives a rib on the second roller to align the endless belt relative to the frame member.
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